Microservice-Based Neural Network Inference

Across Architectural Splits, Cloud Deployment, and Security Hardening

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Context & The Central Problem

Monolithic Status Quo

Most production systems couple model execution into one deployable unit — minimizing overhead but eliminating modularity and explicit security boundaries.

Microservice Promise

Independent deployment, resource placement, and clearer security boundaries — but each split forces intermediate activations across service interfaces, adding serialization and coordination cost.

The Central Problem

Where to split, where to deploy, and how to harden — together, not in isolation. A split acceptable locally may behave very differently in Kubernetes, Azure, and security-hardened contexts.

Research Questions

RQ1 — Architecture & Deployment

- **RQ1.1** Which coarse stage boundaries best decompose ResNet-18?
- **RQ1.2** How does finer-grained refinement affect latency and overhead?
- **RQ1.3** How do activation size and compute distribution explain boundary-crossing behavior?
- **RQ1.4** How does increasing microservice count affect end-to-end cost under local Kubernetes?
- **RQ1.5 / 1.5b** Does the local pattern transfer to AKS, and what is the inter-node penalty?

RQ2 — Security Hardening

- **RQ2.1** What overhead does service identity, managed-Istio mTLS, and authorization policy introduce?
- **RQ2.2** What additional overhead does AMD SEV-SNP confidential VM execution add?
- **RQ2.3** Which hardening configuration offers the best trade-off between cost, complexity, and protection scope?

Approach & Progress

Stage	Method	RQs
1	Local CPU split screening	RQ1.1–1.3
2	Local Kubernetes chain (kind)	RQ1.4
3	Managed AKS chain	RQ1.5, 1.5b
4	mTLS hardening	RQ2.1
5	SEV-SNP confidential execution	RQ2.2
6	Trade-off synthesis	RQ2.3

ResNet-18, FP32, CPU, batch-one fixed throughout. One system layer changed at a time. ACR-pinned image across all AKS stages. Frozen evidence set: parity-validated (max_abs_diff = 0.0) across every condition.

All RQ1 and RQ2 stages complete. Thesis chapters drafted. Currently in writing and revision phase.

RQ1.1–1.3 — Split Selection

Coarse Split Screening (RQ1.1)

Condition	Mean Latency	Overhead
Monolithic	~75.5 ms	—
Split after layer1	—	+4.7% (784 KiB activation)
Split after layer2	80.12 ms	+3.7%
Split after layer3 ✓	79.08 ms	+2.4% (196 KiB)
Split after layer4	—	+3.0% (compute-degenerate)

Early splits penalized by **activation expansion**; late splits by **compute degeneracy**. Carry-forward: split_after_layer3 (main), split_after_layer2 (reference).

RQ1.2 refinement: split_after_layer3_block0 vs. split_after_layer3 — both transfer 196 KiB, but boundary-crossing intervals differ by **7.81 ms** (33.97 vs. 26.16 ms). Compute placement is an independently observable cost dimension.

RQ1.3: Activation transfer dominates at coarse level; compute placement independently observable at refined level. Mid-to-late band is the only region where both dimensions are tolerable.

RQ1.4–1.5b — Deployment Transfer

Local Kubernetes vs. Azure Kubernetes Service

Chain Depth	Local k8s	Local Δ	AKS	AKS Δ
monolithic_k8s_1svc	81.07 ms	—	70.80 ms	—
chain_2svc	83.39 ms	+2.9%	73.44 ms	+3.7%
chain_3svc	87.73 ms	+8.2%	75.41 ms	+6.5%
chain_4svc	90.57 ms	+11.7%	78.29 ms	+10.6%
chain_5svc	92.54 ms	+14.1%	80.52 ms	+13.7%

Ordering Preserved

RQ1.4 ordering transfers to AKS within **1.7 pp** at any condition — triple coverage: loopback → local kind → managed AKS.

Platform-Side Overhead

Non-compute overhead grows from **6.2 ms → 13.6 ms** (chain_2svc → chain_5svc). Added cost is gRPC, kube-proxy, pod network — not compute.

Inter-Node Envelope (RQ1.5b)

Forcing every hop across a node boundary adds **+2.13 ms to +4.09 ms** across chained conditions. Architectural ordering preserved.

RQ2.1 — mTLS, Service Identity & AuthZ

Latency Impact (chain_2svc)

Mean: 75.56 ms → 79.05 ms (**+3.49 ms, +4.62%**). p95: +3.62 ms (+4.53%). Every per-pass delta strictly positive (2.22–5.23 ms). Depth sensitivity at chain_5svc: **+8.84 ms (+10.74%)** — roughly 2.5× the chain_2svc cost.

Ablation: Where the Cost Lives

Plain → mesh-default sidecar: **+2.57 ms** of the +2.84 ms full delta. Strict mTLS and AuthorizationPolicy each within pass-level noise. **Most overhead is entry into the sidecar data path**, not mTLS crypto or policy in isolation.

Operational Footprint

+5 to +14 Kubernetes objects, 34.7–87.6 mCPU sidecar request, 82–224 MiB memory, 6.3–6.8 s schedule-to-ready delta.

Trust Boundary

mTLS terminates at the pod boundary — intermediate activations remain exposed to the runtime. An untrusted partition can reconstruct the input.

RQ2.2 + RQ2.3 — SEV-SNP & Synthesis

SEV-SNP Confidential VM (RQ2.2)

Paired benchmark, mTLS fixed, service2 moved from standard to SEV-SNP node:

Metric	Standard	Confidential
Mean	58.49 ms	60.35 ms (+3.17%)
Median	—	+1.47 ms (+2.52%)
p95	—	+3.60 ms (+5.98%)

Tail amplified **~1.9× relative to mean** — TEE cost is tail-concentrated. SEV-SNP removes cloud host and hypervisor from the TCB, but does not defend against malicious-remote-service activation-inversion threat model.

Synthesis: Composable, Not Substitutable (RQ2.3)

01

mTLS + Service Identity + AuthZ

Modest, symmetric cost (+4.62% mean / +4.53% p95). First-line default.

02

Selective SEV-SNP on Downstream Service

Only when threat model explicitly includes the host platform.

03

Plain AKS

Fastest, but weakest protection.

⚠ Recommendation specific to chain_2svc. mTLS cost grows with depth. Residual trust: mesh control plane, local CA, sidecar-to-app loopback.

Headline Findings, What's Left & Future Work

1. **Coarse split boundaries are not interchangeable**
   Activation expansion penalizes early splits; compute degeneracy penalizes late splits.
2. **Architectural ordering transfers across three substrates**
   Loopback → local k8s → AKS within a 1.7 pp band.
3. **Most chain overhead is platform-side**
   Marginal hop cost depends on the tensor moved at it.
4. **mTLS overhead is bounded but grows with depth**
   Most cost is sidecar data-path entry, not crypto.
5. **SEV-SNP tail amplified ~1.9× relative to mean**
   TEE cost is tail-concentrated, not centered.
6. **Three hardening layers are composable, not substitutable**
   Recommendation depends on threat model and topology.

What's Left

Finalizing analysis prose, cross-RQ synthesis, limitations and threats to validity, front-matter polish.

Future Work

Larger and transformer-based models, GPU execution, activation compression at boundaries, sidecarless / eBPF mesh data planes, adversarial security evaluation against residual CVM attack surfaces.

Peer Feedback & Thank You

Where I'd Like Help

- **Thesis structure** — should RQ1 architectural + deployment collapse into a single strand?
- **Evaluation framing** — is "delta-on-paired-baseline" + "ranking-preserving" defensible?
- **Scope honesty** — am I being too cautious about what the security claims cover?
- **Generalisability** — how strongly should I emphasize ResNet-18, CPU-only, FP32, batch-one specificity?

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